

ADAD AL SHABAB

Senior Data & Operations Analyst

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EDUCATION

United International University, Dhaka
Bachelors - Computer Science and Engineering

2018 – 2023
Dhaka, Bangladesh

Govt. City College, Rajshahi
Higher Secondary School Certificate - Science

2015 – 2017
Rajshahi, Bangladesh

PROFESSIONAL SUMMARY

Senior Data & Operations Analyst with 2.5+ years of software engineering and AI/ML foundation, now applying that depth to **Revenue Cycle Management (RCM)**. Currently leading **Billing Rules Engine team** across 300+ clients and driving measurable reductions in denial rates through automated medical billing. Building internal **infrastructural automations** as well as **running operations**, combining technical fluency with stakeholder management and cross-functional leadership.

EXPERIENCES

[Augmedix BD \(Commure\)](#) [🔗](#) | [Senior Data & Ops Analyst](#)

June 2025 - Present(Full time)

- **Engineer and maintain** the core **RCM billing rule engine**, governing **\$255M+** in claims across **300+** clients — the system has driven a **75% reduction in denial rates** since I took ownership.
- **Leading as primary rule activation authority**, personally **achieved a 90%** same-day (within 24-hour SLA) rule activation rate — an **85% lift in team-wide SLA** adherence.
- **Reduced** incoming ticket inflow from clients by **41% quarter-over-quarter** and **improved claim's First Pass Rate (FPR) by 55%** through proactive rule optimization and **denial root-cause analysis**.
- **Mentored junior analysts** on claims processing with **AR balances, ERA/EOB posting, Denial-Rejection review** and **payer policy documentation**.
- **Spearheaded Quarter-to-quarter internal process automation initiative with 6 internal automations, Retool agents & workflows**, reducing manual rule creation effort up to **58%** alongside: **Rules QA Agent, EHR Rules Agent, Rules Template engine, and Bulk Activation workflows** — adopted org-wide.
- **Leading daily standups** and business **ROI/KPI reporting** with **proficient English communication, stakeholder management**. Creating and maintaining **project roadmaps, WBS, Agile/Scrum-based Jira workflows**.
- **Identified and escalated critical RCM engineering bugs** providing RCA, client-safe workarounds that protect claim accuracy. Managed scope changes and ensuring timely project execution.
- **Identified and Surfaced recurring RCM product gaps**, causing rules to fail to fire on 81% of edited encounters — **directly worked as Forward Deployed Engineer for fix**.

[MelaninMD](#) [🔗](#) | [Software Engineer \(ML\)](#)

Mar 2024 - Mar 2025(Remote)

- **Trained the skin disease classification model** across **46,000+** images spanning **35+** disease types, achieving production-ready inference accuracy.
- **Designed and deployed** a stateless **RESTful inference API** using **Docker, Uvicorn, and AWS EC2**, implementing **structured error handling, JSON serialization**, and multi-part image upload support to serve the model as the backend layer for the **Android/iOS Flutter application**.

- **Engineered** a transactional **PostgreSQL** Stored Procedure pipeline with rollback logic across 35 relational tables — **ensuring 100% data integrity** on failure with **automated error logging** and Root Cause Analysis.
- **Built** Layout-based Segmentation and **OCR pipeline** to extract structured data from invoices, notes & forms, feeding outputs into downstream AI workflows.
- **Contributed** to a **multi-agent orchestration** system using LangGraph, Redis & Pinecone to enable autonomous decision, data processing & task execution.

PROJECTS

Orchestrating Multi Step Agent With LangGraph Memory | Vector DB, Pydantic

Jun 1, 2025

- Enabled **Natural language query parsing**, SQL generation, and dynamic data analysis with memory-augmented context retrieval via **FAISS** and **OpenAI embeddings**.
- Integrated **stateful agent flows**, semantic memory, and SQL toolkits, showcasing advanced use of **LangGraph** for **multi-agent coordination**.

Deep Vision 'Water Ski Sports' AI Model Development | YOLOV8, OpenCV

July 1, 2024

- Goal was to help coaches in deep analysis by integrating custom Computer Vision Model.
- Implemented Real-time features which generated thousands of user interactions.

Diabetes Prediction Advanced ML App | Linear Regression, Streamlit

Feb 10, 2024

- Democratizing advanced Diabetes screening through user-friendly AI web app.
- Helps the medical experts in early detection and prevention of Diabetes.

Advanced Brain Tumor Classification via Conv2D Layer | Deep Learning, Python

Nov 24, 2023

- Used Convolutional Neural Networks to develop a highly accurate and efficient detection system
- DL techniques are implemented to find patterns for accurate detection.

TECHNICAL SKILLS

- Languages:** Python, HTML5, CSS3, Javascript, React.js, Flutter.
- Machine Learning Technologies/Frameworks:** Pandas, Matplotlib, Numpy, TensorFlow, FastApi, Postman.
- LLM Tools/Technologies:** Langchain, n8n, Zapier, Retool, LangGraph, HuggingFace.
- Cloud Development:** Retool, Google Cloud Platform(GCP), AWS, DBT, Streamlit.
- Data Visualization & Analysis Tools:** Google Analytics 4, Excel, Power Bi, Tableau, Selenium, Airtable, Looker Studio.
- Backend/Databases:** SQL, MySQL, PostgreSQL, SQLite, Neo4j (Graph DB)
- Developer Tools:** Jira, ClickUp, Docker, Cursor, Notion, GitHub, VS code, Figma, Canva, Trello, Lucidchart, Confluence.

Coding Platforms

- **Github:** 528+ Contributions (Pro)
- **Kaggle:** Regular Practitioner
- **Streamlit.io:** Deployed 15+ AI/ML Apps
- **HuggingFace Spaces:** Max Rating 1391 (Among the top ML contributors)

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification
 - Statistics - University Of Amsterdam(Coursera)
 - Generative AI Fundamentals - Databricks
 - Database Structures and Management with MySQL - Meta
 - Google Data Analytics Professional Certificate - Google(Coursera)
 - Python For Data Science And AI - IBM
- Aug 17, 2024

Apr 15, 2024

Dec 5, 2023

Nov 27, 2023

Nov 9, 2023

July 4, 2023

EXTRACURRICULAR

- Scientific Article writer on **Medium** [↗](#)
- **Top 10 of 80+ teams at Robi Datathon** [↗](#) — built LLMOPS project to address healthcare problems.
- Top AI App Developer at **Streamlit Cloud** [↗](#)
- LLM AI App Building Contest by **Backdrop Build** [↗](#) - developed OCR based Agentic Co-pilot.
- **Organized** Data Science workshops at **United International University App Forum**.
- **Taught DSA and OS** as Teacher's Assistant at **United International University**.

References

- **Prof. Swakkhwar Shatabda.**
Professor, Department of CSE,
Associate Director, Department of Data Science,
United International University, Dhaka, Bangladesh.